

Bioavailability and Chronic Toxicity of Zinc to Juvenile Rainbow Trout (*Oncorhynchus mykiss*): Comparison with Other Fish Species and Development of a Biotic Ligand Model

KAREL A. C. DE SCHAMPHELAERE* AND COLIN R. JANSSEN

Laboratory of Environmental Toxicology and Aquatic Ecology, Ghent University, J. Plateaustraat 22, B-9000 Gent, Belgium

In this study, the effects of modifying Ca (0.2–4 mM), Mg (0.05–3 mM), Na (0.75–5 mM), and pH (5.5–7.5) on the chronic toxicity of zinc to juvenile rainbow trout (*Oncorhynchus mykiss*) were investigated using standard 30-d assays in which survival and growth were monitored. Survival was observed to be a more sensitive end point than growth, and mortality mainly occurred during the initial stages of the exposure. This suggested that the mode of action of zinc toxicity was mainly of an acute nature. A review and analysis of existing literature demonstrated similar results for most other fish species investigated. Overall, up to a 30-fold variation of zinc toxicity was observed, as indicated by no observed effect concentrations varying between 32.7 and 974 μg of Zn L^{-1} . Increased concentrations of Ca^{2+} , Mg^{2+} , Na^{+} , and H^{+} (within the tested ranges) resulted in a reduction of chronic zinc toxicity by a factor of 12, 3, >2, and 2, respectively. This suggests the major importance of Ca competing with zinc and protecting against zinc toxicity, which seems to be a ubiquitous concept in fish species (and probably also invertebrate). On the basis of the toxicity data obtained, a chronic biotic ligand model (BLM) was developed that takes into account both chemical speciation of zinc and competition between zinc and the above-mentioned cations. The developed model was able to predict chronic effect concentrations with an error of less than a factor of 2 in most cases. Hence, it was concluded that the chronic Zn BLM can reduce toxicity variability due to bioavailability to a considerable extent and that the BLM can become an important tool in criteria setting and risk assessment practice of zinc and zinc substances.

Introduction

Metals may present risks to the aquatic environment and are being managed through regulatory instruments such as environmental quality criteria or risk assessments. It has been recognized by both regulators, industry, and academic scientists that regulatory action should explicitly take into account metal bioavailability since the toxicity of a metal is not only a function of its total or dissolved metal concentration but also of a number of water characteristics, including

dissolved organic carbon (DOC); pH; alkalinity; and concentrations of Ca, Mg, and Na (1–5).

In this context, bioavailability models predicting toxicity have been developed that take into account metal speciation, competition of the toxic metal ion with major cations, and metal binding onto sites of toxic action. In one such model, the sites of toxic action have been termed the biotic ligand, and the associated model was named the biotic ligand model or BLM (see refs 1, 5, and 6 for more chemical, physiological, toxicological, and mathematical background). The BLM integrates most state-of-the-science knowledge on metal bioavailability in a user-friendly computer program that is available from Hydroqual (Mahwah, NJ; via <http://www.hydroqual.com/winblm>).

Initially, the focus of the BLM was on predicting acute toxic effects of metals (i.e., mainly mortality during short exposure periods). Recently, the focus has shifted toward predicting chronic toxic effects (i.e., not only effects on survival but also on growth and reproduction) (7, 8). This is necessary if the BLM is to be used in the EU risk assessment of metals. Indeed, in the EU risk assessment for Zn that is presently ongoing, an agreement was reached on the use of the chronic BLM to take into account bioavailability (9). In previous studies, the chronic Zn BLM for an invertebrate (*Daphnia magna*; our laboratory, unpublished data) and a green alga (*Pseudokirchneriella subcapitata*; 8) were described. In the present study, the development of a chronic Zn BLM for another important taxonomic group is described.

Although a considerable amount of literature data on the chronic toxicity of Zn to freshwater fish species are available (see Table 1; 10–18) and an acute Zn BLM for fish species (rainbow trout, *Oncorhynchus mykiss*, and fathead minnow, *Pimephales promelas*) has been developed (19), not one study has investigated the effects of bioavailability modifying parameters (e.g., hardness or pH) on chronic toxicity of zinc to fish. On the basis of existing literature on Zn uptake (20), acute toxicity, and bioavailability information (3, 19), we selected Ca^{2+} , Mg^{2+} , Na^{+} , and H^{+} (pH) as potentially important factors affecting chronic Zn toxicity to rainbow trout. It was hypothesized, in accordance with the cation competition concept of the BLM, that increased concentrations of these cations would result in a reduction of chronic Zn toxicity and that these effects could be modeled using the BLM concept. To test this hypothesis, juvenile rainbow trout were exposed to Zn in media containing different concentrations of these cations, and the effects on survival and growth were monitored during a 30-d exposure.

Experimental Section

Fish. Juvenile rainbow trout (28–35 d post-hatch, swim fry, fully resorbed yolk sac) were purchased from Houghton Springs Fish Farm (United Kingdom), where they are hatched and cultured in freshwater with pH ~7.5, hardness ~50–70 mg of CaCO_3/L , and dissolved Zn ~5 $\mu\text{g}/\text{L}$ (long-term averages, Hans Hoff, Houghton Springs Fish Farm, UK, personal communication). Upon arrival in our laboratory, the fish were acclimated for 1 week to control exposure media (dissolved Zn <4.6 $\mu\text{g}/\text{L}$) under the same conditions as in the exposures (see below). For each test, eight rainbow trout were held in each of the 18 polyethylene aquaria containing 3 L of control test medium (no zinc added). Mortality during this acclimation period was less than 5% in all cases, which is in accordance with the guideline for testing chemicals (21).

Experimental Design. The effects of bioavailability modifying factors on the chronic (30 d) toxicity of Zn to rainbow trout were examined in five series of assays. The first series,

* Corresponding author telephone: +32 9 264 79 32; fax: +32 9 264 37 66; e-mail: Karel.Deschampelaere@Ugent.be.

TABLE 1. Summary of Effects of Chronic Exposures to Zinc for Different Life Stages of Seven Fish Species As Measured by Different Ecologically Relevant End Points (as μg of Zn/L)

species ^a	life stages tested ^b	t_e ^c	NOEC _s ^d	LOEC _s ^e	other end points tested ^f	t_x ^g	NOEC _x ^h	LOEC _x ⁱ	mortality at LOEC _s (%)	effect on end point x at LOEC _s (%)	mortality at LOEC _x (%)	effect on end point x at LOEC _x (%)	SSS ^j	C/A ^k	stabilized LC ₅₀ (d)	ref
brook trout	1, 2, <u>3</u> , 4 ^m	8 w	716	1,368	r, g, (h) ⁿ	5 m	534	1360	44	38	— ^o	— ^o	0.75	NR ^p	NR	10
fathead minnow																
5- or 7-d test	2	5–7 d	117–291	285–615	g	5–7 d	117–291	270–615	NR	NR	NR	NR	0.36 ^q	0.8–0.1	4–7	11
7-d test	2	7 d	84.6	184	g	7 d	184	374	<50 ^r	13	>50 ^r	50	1	1	4	12
32-d test	1	32 d	129	275	g	32 d	—	>275	NR	NR	NR	NR	1	NR	NR	11
life cycle	1, 2 ^s	8 w	145	295	(r), h	LC ^t	78	145	54	92	4	79	0	NR	~14 ^u	13
flagfish	1, <u>2</u>	30 d	51	85	r, (g), h	100 d	26	51	79	16	13	12	0.5	NR	NR	14
humble minnow	4 ^v	150 d	50	130	g	150 d	50	130	20	<10	— ^o	— ^o	0.75	NR	NR	15
mottled sculpin	2–3 ^w	30 d	16	27	g	30 d	—	>27 ^x	32	6	— ^x	— ^x	1	0.2	9–13	16
rainbow trout																
this study ^y	3	30 d	31.5–401	78.9–2310	g	30 d	159–974	351–3580	see summary of effects in results section				0.93 ^z	0.6–1	7–14	
soft water	1, 3	25 d	36	71	g, h	2 y	—	>547	8%	NR	NR	NR	1	NR	5 ^{aa}	17
hard water	4	21 m	320	640	r, g	21 m	—	>2200	6.4%	NR	NR	NR	1	NR	NR	17
zebrafish ^{bb}	1	NA ^{cc}	6210	12 400	h	NA ^{cc}	1280	2100	NR	NR	NR	NR	NA ^{cc}	NR	NR	18

^a Scientific names: brook trout, *Salvelinus fontinalis*; fathead minnow, *Pimephales promelas*; flagfish, *Jordanella floridae*; humble minnow, *Phoxinus phoxinus*; mottled sculpin, *Cottus bairdii*; rainbow trout, *Oncorhynchus mykiss*; zebrafish, *Danio rerio*. ^b Numbers refer to life stage at start of exposure: 1, embryo/egg; 2, sac fry; 3, swim fry; 4, older organisms. Underlined life stage is the most sensitive life stage for survival if other life stages were also reported. ^c t_e , exposure period associated with reported NOEC_s; d = days; w = weeks; m = months; y = years. ^d No observed effect concentration for survival. ^e Lowest observed effect concentration for survival. ^f Other end points tested in same study: r = reproduction, g = growth, h = hatchability. End point in parentheses is most sensitive end point besides survival if more than one end point was reported; this end point is referred to as "x" in other column titles. ^g t_x , exposure period associated with reported NOEC_x; d = days, w = weeks, m = months, y = years. ^h NOEC_x, no observed effect concentration for most sensitive end point other than survival. ⁱ LOEC_x, lowest observed effect concentration for most sensitive end point other than survival. ^j Survival Sensitivity Score for assessing sensitivity of survival compared to other end points. SSS = (LOEC-based score + concentration-response score)/2. The LOEC-based score is obtained as follows: 1 point if LOEC_s < LOEC_x, 0.5 point if LOEC_s = LOEC_x, 0 points if LOEC_s > LOEC_x. The concentration response score is obtained by comparing effects on survival (% mortality, m) and on other end points (% effect compared to control, e). First, the lowest LOEC is considered and 1 point is given if $m > e$ and 0 points if $m < e$. In cases where this cannot be clearly determined, the effects at the highest LOEC are compared (i.e., 1 point if $m > e$, 0 points if $m < e$). If SSS > 0.5, survival is most likely the most sensitive end point. ^k Chronic LC₅₀/acute LC₅₀, acute LC₅₀ = 96-h LC₅₀. ^l Indicates the exposure duration where mortality has ceased or where no further mortality was noted for the rest of the exposure period. ^m Life stage 4 = 70 g fingerlings; most sensitive life stage based on highest mortality at LOEC_s. ⁿ Hatchability of eggs spawned from first generation. ^o LOEC_s equal to LOEC_x, same values as in previous two columns apply. ^p NR, data not reported in original papers or essential data missing to calculate. ^q Mean score for a total of 7 tests. ^r Based on comparison with acute 96-h LC₅₀. ^s Based on mortality at 295 μg of Zn/L, 54% for eggs, 84% for sac fry. ^t Life cycle exposure, exact exposure duration not reported. ^u All deaths occurred in first 14 d of exposure. ^v Mature yearlings. ^w Exact life stage not reported, "recently emerged" larvae of ~0.5 g. ^x All fish died at the next tested concentration; 53 μg of Zn/L and growth could not be determined. ^y This study, see Table 3 for detailed reporting of effects. ^z In 14 out of 15 tests, survival was more sensitive than growth (see also in text). ^{aa} Based on observation that no fish died beyond 5th day of exposure. ^{bb} Mean effective hatching time and mean effective survival time were reported; effect concentrations not based on percent effect. ^{cc} NA, not applicable. Exposures started with eggs in the blastula stage and continued to 90% mortality of the hatched larvae. Significant reduction of hatching time occurred at lower concentrations than reduction of survival time. However, the end point survival time integrates the end points hatching; survival of eggs and survival of hatched larvae. Since the authors suggested that delayed hatching may be protective against zinc-induced mortality (18) and since the percentage of eggs that eventually hatched was not reported, it cannot unambiguously be concluded here that hatching is a more sensitive an end point than survival or vice versa.

TABLE 2. Physicochemical Characteristics of Test Media in Which Toxicity Tests with Rainbow Trout Were Conducted^a

code ^b	pH	Ca (mM)	Mg (mM)	Na (mM)	K (mM)	SO ₄ (mM)	Cl (mM)	DIC (mM) ^c
RF-B	7.45 (0.02)	0.221 (0.012)	0.075 (0.011)	0.773 (0.076)	0.056 (0.004)	0.123 (0.001)	0.514 (0.024)	0.974 (0.087)
RF-CA4	7.54 (0.02)	3.955 (0.041)	0.063 (0.002)	0.854 (0.052)	0.049 (0.003)	0.123 (0.001)	8.825 (0.211)	0.834 (0.018)
RF-MG3	7.39 (0.02)	0.197 (0.005)	3.121 (0.061)	0.822 (0.066)	0.050 (0.001)	0.120 (0.002)	6.479 (0.413)	0.916 (0.090)
RF-NA5	7.49 (0.02)	0.228 (0.020)	0.072 (0.004)	4.655 (0.133)	0.052 (0.008)	0.122 (0.003)	5.020 (0.113)	0.926 (0.052)
CA1-B	7.75 (0.04)	0.235	0.067	0.723	0.052	0.118	0.403	0.876
CA1-1	7.80 (0.02)	0.986	0.068	0.758	0.050	0.120	1.841	0.975
CA1-2	7.87 (0.04)	1.844 (0.126)	0.057 (0.003)	0.681 (0.071)	0.039 (0.008)	0.128 (0.001)	4.073 (0.049)	1.368 (0.137)
CA1-4	7.85 (0.05)	3.917	0.065	0.749	0.048	0.120	7.711	0.952
CA2-B	7.61 (0.03)	0.228 (0.014)	0.063 (0.007)	0.794 (0.098)	0.054 (0.008)	0.116 (0.002)	0.456 (0.099)	0.991 (0.082)
CA2-1	7.58 (0.03)	0.953 (0.026)	0.068 (0.004)	0.740 (0.036)	0.049 (0.007)	0.122 (0.003)	1.948 (0.170)	1.107 (0.060)
CA2-4	7.68 (0.03)	3.886 (0.078)	0.069 (0.002)	0.742 (0.062)	0.051 (0.003)	0.116 (0.002)	7.608 (0.241)	0.922 (0.044)
MG-B	7.65 (0.02)	0.214 (0.016)	0.078 (0.003)	0.789 (0.053)	0.049 (0.005)	0.121 (0.003)	0.569 (0.030)	0.907 (0.107)
MG-0.2	7.73 (0.02)	0.224 (0.020)	0.227 (0.018)	0.731 (0.030)	0.047 (0.006)	0.125 (0.001)	0.932 (0.038)	0.853 (0.093)
MG-1	7.74 (0.03)	0.241 (0.020)	1.152 (0.110)	0.919 (0.130)	0.049 (0.008)	0.128 (0.001)	2.901 (0.105)	0.933 (0.103)
MG-2	7.79 (0.02)	0.229 (0.015)	2.060 (0.137)	0.773 (0.056)	0.050 (0.006)	0.124 (0.004)	5.289 (0.338)	0.946 (0.075)
MG-3	7.65 (0.02)	0.210 (0.009)	3.123 (0.065)	0.713 (0.004)	0.050 (0.005)	0.123 (0.003)	7.443 (0.209)	0.836 (0.067)
PH-5.5	5.68 (0.06)	0.218 (0.007)	0.073 (0.009)	4.509 (0.239)	0.041 (0.004)	0.119 (0.006)	6.461 (0.174)	0.294 (0.022)
PH-6.5	6.70 (0.02)	0.223 (0.009)	0.064 (0.002)	4.522 (0.184)	0.048 (0.009)	0.119 (0.003)	6.130 (0.305)	0.599 (0.043)
PH-7.5	7.58 (0.02)	0.221 (0.013)	0.063 (0.030)	4.858 (0.231)	0.049 (0.006)	0.122 (0.003)	5.436 (0.465)	0.806 (0.022)

^a Numbers in parentheses indicate the standard error. ^b Test codes are explained in section on experimental design. ^c Dissolved inorganic carbon.

TABLE 3. Acute and Chronic Effect Concentrations (μg of Zn L⁻¹) and Growth Data for Rainbow Trout Exposed to Zinc for 30 d^a

code	96-h LC ₅₀ ^b	30-d LC ₅₀	30-d LC ₁₀ ^c	30-d NOEC _s ^d	30-d LOEC _s ^e	r ₃ control (d ⁻¹)	r ₃ at LOEC _s (d ⁻¹)	NOEC _g ^f	LOEC _g ^g
RF-B	209 (154–283)	177 (132–238)	38.4 (19.7–74.5)	31.5	117	0.0419 (0.0060)	0.0518 (0.0039)	—	>375
RF-CA4 ^h	NC ⁱ	NC	NC	—	>1530	0.0532 (0.0029)	—	—	>1530
RF-MG3	730 (664–800)	583 (504–676)	NC	90.3	538	0.0464 (0.0039)	0.0509 (0.0075)	—	>538
RF-NA5	441 (277–700)	282 (196–450)	83.2 (24.9–278.1)	95.1	371	0.0422 (0.0029)	0.0472	—	>371
CA1-B	171 (144–204)	NP ^j	NP	NP	NP	NP	NP	NP	NP
CA1-1	1170	NP	NP	NP	NP	NP	NP	NP	NP
CA1-2	1470 (910–2300)	1240 (760–2040)	290 (96–880)	974	2310	0.0483 (0.0063)	0.0350 (0.0090)	974	2310
CA1-4	2560 (1670–3940)	NP	NP	NP	NP	NP	NP	NP	NP
CA2-B	194 (152–248)	159 (110–228)	34.5 (14.5–82.4)	78.9	166	0.0434 (0.0020)	0.0284 (0.0073)	166	345
CA2-1	904 (608–1343)	759 (568–1014)	171 (91–324)	169	363	0.0358 (0.0026)	0.0280 (0.0068)	—	>3580
CA2-4	2280 (1460–3560)	1860 (1230–2820)	337 (113–1010)	786	1670	0.0391 (0.0044)	0.0301 (0.0069)	—	>1670
MG-B	130 (100–168)	108 (81–143)	46.1 (25.5–83.6)	48.0	162	0.0266 (0.0048)	0.0502 (0.0039)	—	>1380
MG-0.2	153 (113–208)	151 (121–189)	67.9 (33.4–138.0)	45.4	173	0.0280 (0.0058)	0.0237 (0.0144)	—	>1410
MG-1	214 (176–261)	212 (183–247)	103 (77–139)	151	338	0.0265 (0.0054)	0.0239 (0.0081)	—	>1280
MG-2	283 (208–384)	224 (185–247)	104 (70–156)	159	351	0.0301 (0.0037)	0.0101 (0.0101)	159	351
MG-3	483 (427–547)	339 (269–428)	99.1 (58.7–167.5)	165	356	0.0274 (0.0078)	0.0300 (0.0087)	—	>1020
PH-5.5	1510 (1200–1910)	802 (630–1023)	312 (184–531)	401	800	0.0362 (0.0038)	0.0296 (0.0070)	—	>800
PH-6.5	548 (365–822)	368 (214–632)	99.1 (30.3–324.3)	256	665	0.0349 (0.0065)	0.0507 (0.0022)	—	>1280
PH-7.5	610 (398–931)	361 (226–579)	73.6 (23.4–231.7)	157	256	0.0286 (0.0032)	0.0264 (0.0075)	—	>1740

^a Growth data are given as pseudo-specific growth rate (r₃) calculated according to eq 1. Numbers in parentheses indicate 95% confidence interval for effect concentrations and standard error of mean for growth rate. ^b LC₅₀, median lethal concentration. ^c LC₁₀, lethal concentration for 10% of the organisms. ^d No observed effect concentration for end point survival. ^e Lowest observed effect concentration for end point survival. ^f No observed effect concentration for end point growth. ^g Lowest observed effect concentration for end point growth. ^h No significant effects on growth or survival were observed at the highest concentration of 1530 μg of Zn L⁻¹. ⁱ NC, could not be calculated. ^j NP, test not performed.

a range finding test (Code RF in Tables 2 and 3), was conducted to assess the possible range of protective effects of Ca, Mg, and Na on acute and chronic zinc toxicity. One test was conducted with a water with low concentration of all ions (0.2 mM Ca, 0.05 mM Mg, 0.75 mM Na; code RF-B), and three tests were performed in which the concentration of one cation was increased (i.e., 4 mM Ca, 3 mM Mg, or 5 mM Na; codes RF-CA4, RF-MG3, or RF-NA5, respectively).

The results of these tests were used to choose zinc concentrations to be tested in the subsequent four test series. Two were carried out to investigate the individual effects of Ca. In the first series 0.2, 1, 2, and 4 mM were tested (codes CA1-B, CA1-1, CA1-2, and CA1-4); in the second series 0.2, 1, and 4 mM were tested (codes CA2-B, CA2-2, and CA2-4). In a third series, tests were conducted at Mg concentrations of 0.05, 0.2, 1, 2, and 3 mM (codes MG-B, MG-0.2, MG-1, MG-2, and MG-3, respectively). In the final test series, three

pH levels were tested (i.e., 5.5, 6.5, and 7.5; codes PH-5.5, PH-6.5, and PH-7.5, respectively). The effect of Na was not further investigated as the range finding test (RF-NA5) had demonstrated its relatively low importance as compared to other effects (see Results and Discussion section). It was anticipated that this design would allow the development of a chronic Zn BLM using the approach of De Schampelaere and Janssen (1), provided that the observed effects were in accordance with the concept of cation competition, one of the key assumptions of the BLM.

Exposures. All test media were prepared using carbon-filtered, deionized water (conductivity <2 $\mu\text{S cm}^{-1}$, DOC ~300 $\mu\text{g L}^{-1}$) and using chemicals (p.a. grade) purchased from VWR International (Leuven, Belgium). The acclimated rainbow trout (35–42 d old, wet wt ~200 mg) were exposed for 30 d in a gravitational flow-through system with a renewal rate of 2 L (g of fish)⁻¹ d⁻¹. At the start of each test, a random

sample of 10 animals was taken for determination of the initial weight of the fish. For each test medium, 4 or 5 zinc concentrations (spiked as ZnCl_2) and a control were tested. All test media were equilibrated for 1 d before use. Each concentration was tested in triplicate in polyethylene aquaria containing 3 L of test medium and holding five animals each. At all times water temperature was within 1 °C of the target temperature of 15 °C. Aquaria were vigorously aerated, and a light cycle of 12 L:12 D was used. Each day fish were fed 4% of their wet body weight with a commercial fish food (i.e., wet weight of food/average wet body weight of fish at start of test = 0.04), and this ration was increased to 6% after 14 d of exposure. Uneaten food and faeces were siphoned from the bottom of the aquaria three times a week. Each day mortality was noted, and dead fish were removed. At the end of the exposure (i.e., after 30 d), surviving fish were weighed to the nearest 0.1 mg.

Chemical Analyses. At regular intervals a number of physicochemical parameters were determined. pH, water temperature, and oxygen saturation were measured daily. NO_2^- and NO_3^- were monitored at least once a week using spectrophotometric test kits (Merck, Darmstadt, Germany). The following parameters were measured twice a week after filtration through 0.45 μm filters (Gelman Sciences, Ann Arbor, MI). Dissolved organic carbon (DOC) and dissolved inorganic carbon (DIC) were measured with a TOC analyzer (TOC-5000, Shimadzu, Duisburg, Germany). Cl^- and SO_4^{2-} were measured using ion chromatography (DIONEX 2000i/SP, Dionex, Sunnyvale, CA), Ca, Mg, Na, and K using inductively coupled plasma spectrophotometry (ICP, Spectro Analytical Instruments, Kleve, Germany). For DOC, DIC, Cl^- , SO_4^{2-} , Ca, Mg, Na, and K samples from all zinc treatments and all replicates were pooled. Total and dissolved Zn concentrations were measured twice a week using flame atomic absorption spectrophotometry (SpectraAA100, Varian, Mulgrave, Australia) after acidification of the samples with 0.14 N HNO_3 . Samples from the three replicate aquaria were pooled. Calibration standards (Sigma-Aldrich, Steinheim, Germany) and a reagent blank were analyzed with every 10 samples. The detection limit for Zn was always $<4.6 \mu\text{g/L}$. Two certified reference samples, TMDA-62 and TM-25.2 (National Water Research Institute, Burlington, ON, Canada), with certified Zn concentrations (mean $\pm$ 95% confidence interval) of $110 \pm 15.5 \mu\text{g L}^{-1}$ and $24 \pm 4.6 \mu\text{g L}^{-1}$, respectively, were analyzed at the beginning and end of each series of Zn measurements. Measured values were always within 10% of the certified value.

Data Treatment, Statistics, and Reporting. Using body weights at the start and end of each test, pseudo-specific growth rates were calculated for each surviving fish according to following equation (21):

$$r_3 = \frac{\ln W_2 - \overline{\ln W_1}}{t_2 - t_1} \quad (1)$$

with r_3 as the pseudo-specific growth rate (d^{-1}); W_1 and W_2 as the weight of fish at t_1 (0 d) and t_2 (30 d), respectively; $\ln W_2$ is the natural logarithm of the weight of an individual fish at t_2 ; $\overline{\ln W_1}$ is the average of the natural logarithms of W_1 of the 10 randomly selected fish measured at t_1 ; t_1 and t_2 are the time (d) at the start and end of the test, respectively.

Growth rate data in zinc exposures were compared to those in the control using a two-sided Student's t -test for independent samples to test if growth rates were different from the control growth rate. Mortality was statistically compared to the control using analysis of variance, followed by a one-sided Dunnett test to test if mortality was higher than in the control. Differences in growth or mortality were considered significant at $p < 0.05$. These analyses were used

to derive the no observed effect concentration (NOEC) and the lowest observed effect concentration (LOEC) for growth (NOEC_g , LOEC_g) and survival (NOEC_s , LOEC_s). NOEC and LOEC values taken from literature are noted with the subscript "r" or "h" to indicate the use of reproduction or hatchability end points, respectively.

Survival in each replicate aquarium was plotted against average measured dissolved zinc concentrations during the tests. A logistic response model was fitted to the data (22):

$$y = \frac{1}{1 + \left(\frac{x}{\exp(a)}\right)^s} \quad (2)$$

with y as the fraction of organisms that survived the exposure (between 0 and 1); $a = \ln(\text{LC}_{50}) = \log_e(\text{LC}_{50})$; s is the slope parameter; x is the average measured dissolved Zn concentration; and LC_{50} is the median lethal concentration. This model was applied to both 96-h and 30-d survival data. For 30-d survival data, the equation was re-parameterized to explicitly have the LC_{10} (lethal concentration for 10% of the organisms) as one of the parameters to be estimated, allowing an easy estimation of confidence intervals directly from the fitting procedure, as suggested by Bruce and Versteeg (23):

$$y = \frac{1}{1 + \left(\frac{x}{\exp(a)}\right)^{(\ln(1/9)/b-a)}} \quad (3)$$

with $b = \log_e(\text{LC}_{10}) = \ln(\text{LC}_{10})$. Parameter estimation and calculation of the 95% confidence limits was carried out using the Levenberg–Marquardt method (24, 25). All statistics, hypothesis testing, and curve fittings were carried out using Statistica 6.0 software (Statsoft, Tulsa, OK).

Results and Discussion

Water Quality. Detailed water chemistry during all experiments is given in Table 2. Frequent measurements during the first test series (range finding test) yielded the following information with regard to the general water quality during the flow-through tests: oxygen saturation was always higher than 80%, but for >90% of the measurements it was >90%. NO_3^- and NO_2^- were always $<0.1 \text{ mg L}^{-1}$, DOC levels were generally below 1 mg L^{-1} ; the maximum value of DOC recorded was 1.4 mg L^{-1} . The background DOC level of the deionized dilution water is $\sim 0.3 \text{ mg of DOC L}^{-1}$ (1). These measurements indicate an overall good performance of the flow-through system and the good water quality throughout the test. Very constant levels of cations (Ca, Mg, Na, K) and anions (Cl^- , SO_4^{2-}) were observed during the range-finding test. Measured levels were mostly within 10% of the nominal values. Hence, the frequency of all measurements (except Zn) was reduced in the subsequent tests: DOC, DIC, Ca, Mg, Na, K, Cl^- , and SO_4^{2-} were measured once instead of twice a week, and pH (except for the pH test), oxygen saturation, and temperature were measured twice a week instead of daily. Zn concentrations in the control medium, as measured during the tests, were always below the detection limit of the flame AAS ($4.6 \mu\text{g}$ of Zn L^{-1}).

Effects of Zinc on Survival and Growth. The effect concentrations of Zn based on survival and growth of the juvenile rainbow trout are summarized in Table 3. Pseudo-specific growth rates of juvenile rainbow trout in the control exposures were between 0.0265 and 0.0532 d^{-1} (Table 3). The growth rate of fish at Zn concentrations up to the LOEC_s was not significantly lower than the control growth rate in 12 out of 14 tests. In 5 out of 14 test growth rates of zinc-exposed fish at the LOEC_s were even higher than those of nonexposed fish. Significant reduction of growth only oc-

TABLE 4. Biotic Ligand Model^a Constants for Acute and Chronic Zn Exposures of Rainbow Trout (*Oncorhynchus mykiss*) and *Daphnia magna* and Stability Constants for Inorganic Zinc Complexes

ref	<i>Daphnia magna</i>	rainbow trout				stability constants for inorganic zinc species	
	acute (48 h)	acute (96 h)	acute (96 h)	chronic (30 d)	zinc	log <i>K</i> (26)	
	(3)	(19)	this study	this study			
log <i>K</i> _{CaBL}	3.3	4.8 ^b	(Cd – 4.5)	3.8	3.6	ZnOH ⁺	5.00
log <i>K</i> _{MgBL}	3.1	– ^c	(Cd – 3.5)	3.5	3.1	Zn(OH) ₂	10.2
log <i>K</i> _{NaBL}	2.4	– ^c	(Cd – 3.0)	2.9	2.4	ZnHCO ₃ ⁺	11.83
log <i>K</i> _{HBL}	<6 ^d	6.7 ^e	(Cd – 6.7)	6.7	6.3	ZnCO ₃	4.76
log <i>K</i> _{ZnBL}	5.3	5.5	(Cd – 8.6)	5.5 ^f	5.5 ^f	ZnSO ₄	2.34
LA ₁₀ ^h	NR ⁱ	NR	NR	ND ^j	0.77 ± 0.31 ^k	ZnCl ⁺	0.4
LA ₅₀ ^h	3.46	0.63–2.29	NR	1.73 ± 0.64	2.04 ± 0.50		

^aUsing these constants with the BLM software (BLM Windows version 1.0.0., Hydroqual, Mahwah, NJ) LC₁₀ and/or LC₅₀ values can be predicted on a dissolved Zn basis. Gill-binding constants for Cd reported in ref 19 are added for comparison. ^bFitted on toxicity data. ^cNot taken into account in this version of the BLM. ^dInitially not reported, but <6 represents no significant competition at lowest tested pH (i.e., 6). ^eAssumed equal to constant obtained by Playle et al. (32) from a Cd-gill binding study with fathead minnows. ^fAssumed equal as in Santore et al. (19), based on data of Alsop and Wood (28). ^hAs nmol g^{–1} wet weight. ⁱNR, not reported. NOEC instead of LC₁₀. ^jND, not determined. ^kMean ± 95% confidence interval.

curred at concentrations at which significant mortality was observed. Additionally, in 25 out of 27 treatments (different test medium and different Zn concentration) where significant effects on survival were detected, the percent mortality was larger than the percent reduction of growth rate (data not shown). From all this, it can be concluded that survival is a more sensitive end point than growth in 30-d assays with juvenile rainbow trout.

Also important is the fact that only a small difference between 96-h and 30-d mortality data was observed. The ratio between 30-d and 96-h LC₅₀ was between 0.59 and 0.99, indicating that mortality mostly occurred during the first 4 d of the exposure. Moreover, in all exposures mortality ceased between day 7 and day 14 (data not shown). These data suggest a mainly acute mode of toxic action of zinc on juvenile rainbow trout.

Effects of Physicochemistry on Acute and Chronic Toxicity of Zinc. As no concentration–response curves could be established for growth rate (see above), it was not possible to estimate effect concentrations for growth, a requirement for deriving BLM constants according to the method of De Schampheleere and Janssen (1). Therefore, only effect concentrations based on survival are discussed and used for the BLM development. Overall, 96-h LC₅₀ values varied between 130 and 2830 µg of Zn L^{–1} (factor 22 difference), 30-d LC₅₀ values between 108 and 1860 µg of Zn L^{–1} (factor 17), 30-d LC₁₀ between 34.5 and 337 µg of Zn L^{–1} (factor 10), and 30-d NOECs between 32.7 and 974 µg of Zn L^{–1} (factor 30). This factor 10–30 variability in effect concentrations clearly demonstrates the importance of taking bioavailability of zinc into account in regulatory risk assessments.

First, range finding tests (test code RF) revealed that Ca, Mg, and Na all reduced chronic zinc toxicity. On the basis of 30-d LC₅₀ values, increased concentrations of Ca (4 mM, RF-CA4), Mg (3 mM, RF-MG3), and Na (5 mM, RF-NA5) reduced Zn toxicity by a factor of >8, 3, and somewhat less than 2, respectively, when compared to the standard test medium, which is low in Ca, Mg, and Na (Table 3). This suggests the following order of importance: Ca > Mg > Na. The results for RF-CA4 could not be used quantitatively as no significant mortality was observed, even at a zinc concentration as high as 1530 µg L^{–1}. The results of the RF-MG3 test were not used to derive the stability constant for Mg but were taken into account for the estimation of the mean LA₁₀ or LA₅₀ (the “lethal accumulation” of zinc on the biotic ligand that is associated with 10% and 50% mortality, respectively, see further). The results of the RF-NA5 test were used together with the RF-B test for the estimation of a

stability constant for Na competition. The latter constant should be considered a relatively crude estimate; hence, the resulting model should be employed with caution for toxicity predictions in waters with Na concentrations higher than 5 mM.

Given the importance of Ca and Mg, additional testing with these cations was subsequently performed (i.e., from 0.2 to 4 mM Ca and from 0.05 to 3 mM Mg). The results of these univariate experiments are also reported in Table 3. Both Ca test series revealed a decreased Zn toxicity (both acute and chronic) at increasing Ca concentrations. In the first test series (code CA1), 96-h LC₅₀ values increased from 171 to 2560 µg of Zn L^{–1} with an increase of Ca from 0.2 to 4 mM (factor 15); in the second test series (code CA2), 96-h LC₅₀ increased from 194 to 2280 µg of Zn L^{–1} (factor 12). The 30-d LC₅₀ increased from 151 to 1860 µg of Zn L^{–1} (factor 12). The Mg test series (code MG) revealed that an increase of Mg from 0.05 to 3 mM yielded a decrease of toxicity by a factor of 3 for chronic toxicity (30-d LC₅₀ values between 108 and 339 µg of Zn L^{–1}) and a factor of 4 for acute toxicity (96-h LC₅₀ values between 130 and 483 µg of Zn L^{–1}). Thus, the higher importance of Ca versus Mg was confirmed.

The effect of pH is also presented in Table 3. At pH 5.7, the 30-d LC₅₀ was 802 µg of Zn L^{–1}, which is more than a factor of 2 higher than the 30-d LC₅₀ at pH 6.7 (368 µg of Zn L^{–1}) and at pH 7.5 (361 µg of Zn L^{–1}). This suggests a competition effect between free zinc ions and H⁺ ions.

In summary, the effect of the investigated physicochemical parameters on zinc toxicity can be ranked in order of decreasing importance (factor effect based on 30-d LC₅₀ values in parentheses): Ca (12) > Mg (3) ~ pH (>2) > Na (2).

Development of the Chronic Zn BLM. Speciation calculations of the 96-h and 30-d LC₅₀ values were carried out using the BLM software (Windows version 1.0.0; Hydroqual, Mahwah, NJ) and using constants for inorganic zinc complexes taken from Martell et al. (26) and summarized in Table 4. During the tests, DOC concentrations in 92% of the samples were below 1 mg L^{–1} with a maximum of 1.4 mg L^{–1}. To estimate the potential influence of DOC concentrations on the variation of Zn speciation during the tests, calculations were performed at 0 mg of DOC L^{–1} (no significant organic complexation assumed of the background DOC present in deionized water used to make up the test media, i.e., 0.3 mg L^{–1}) and at 1 mg of DOC L^{–1} (100% fulvic acid).

The results of the speciation calculations are given in Figure 1, where the 96-h and 30-d LC₅₀ values (as Zn²⁺ activity) as a function of the chemical activity of a competing cation

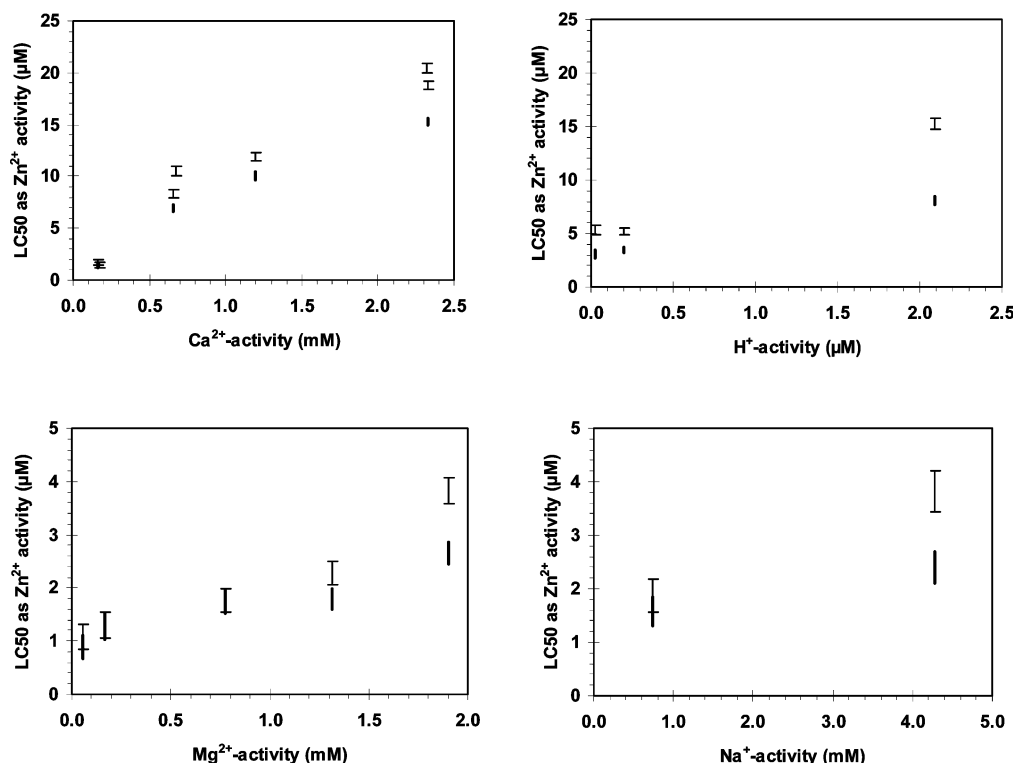


FIGURE 1. Effect of Ca^{2+} , Mg^{2+} , H^{+} , and Na^{+} on acute (96 h, thin bars) and chronic (30 d, thick bars) LC_{50} values of Zn expressed as Zn^{2+} activity. Results are presented as intervals, with the top of each bar being the $\text{LC}_{50 \text{ Zn}^{2+}}$ calculated using no dissolved organic carbon (DOC) as assumption, and the bottom of each bar using 1 mg DOC L^{-1} (100% as fulvic acid, see text for further details). These data were used for the derivation of the BLM constants (Table 4).

are presented. Complexation of Zn to organic matter does not seem to be very significant under the conditions during the toxicity tests. On average only 19% of the zinc was complexed to organic matter if 1 mg of DOC L^{-1} was assumed. Free Zn^{2+} activities were always less than a factor of 1.5 lower at 1 mg of DOC L^{-1} than at 0 mg of DOC L^{-1} (Figure 1).

Figure 1 indicates that LC_{50} values (as Zn^{2+} activity) are higher when the chemical activity of the cations Ca^{2+} , Mg^{2+} , Na^{+} , and H^{+} is increased. This is in line with the BLM concept of cation competition (1, 3). On the basis of these data, stability constants for competition of Ca^{2+} ($\log K_{\text{CaBL}}$), Mg^{2+} ($\log K_{\text{MgBL}}$), Na^{+} ($\log K_{\text{NaBL}}$), and H^{+} ($\log K_{\text{HBL}}$) were estimated according to the method explained in De Schampelaere and Janssen (1).

Following values of constants were obtained (Table 4): $\log K_{\text{CaBL}} = 3.8$, $\log K_{\text{MgBL}} = 3.5$, $\log K_{\text{NaBL}} = 2.9$, and $\log K_{\text{HBL}} = 6.7$ for the 96-h exposure and $\log K_{\text{CaBL}} = 3.6$, $\log K_{\text{MgBL}} = 3.1$, $\log K_{\text{NaBL}} = 2.4$, and $\log K_{\text{HBL}} = 6.3$ for the 30-d exposure. The slightly lower competition constants obtained with chronic versus acute Zn toxicity results (about 0.2–0.5 log units) are due to the lesser increase of 30-d LC_{50} values with increasing cation activities for chronic toxicity (Figure 1). This corroborates the findings reported for *D. magna* in which competition constants also tend to be lower in chronic exposures (Heijerick et al., unpublished data). Similarly, Hogstrand et al. (27) have observed that the affinity of the apical Ca channel in fish gills decreases upon chronic sublethal zinc exposure.

The constants mentioned above and summarized in Table 4 were obtained using the speciation calculation results based on the assumption of 0 mg of DOC L^{-1} in the test media. When assuming 1 mg of DOC L^{-1} , all stability constants were only slightly higher (~ 0.3 log units, not shown). However, higher DOC concentrations in test aquaria were only measured in the final stages of the assays whereas the effects of zinc were determined in the early stages of the tests (see

above). Hence, the constants derived for 0 mg of DOC L^{-1} are probably more realistic. These constants will be considered in the final BLM and in all comparisons below.

Once the competition constants are derived, some additional parameters are needed in the BLM: the stability constant for Zn binding to the biotic ligand (K_{ZnBL}), the density of Zn binding sites on the gill (B_{max}), and the lethal accumulation of zinc on the fish gill resulting in a given effect level (e.g., at 10% and 50% mortality after 30-d exposure this is termed the LA_{10} and the LA_{50} , respectively). When K_{ZnBL} and B_{max} are known, 96-h LA_{50} , 30-d LA_{10} , and 30-d LA_{50} values can be calculated for each test with the BLM software (Windows version 1.0.0; Hydroqual, Mahwah, NJ) using the 96-h LC_{50} , 30-d LC_{50} , or 30-d LC_{10} values (Table 3), the water chemistry (Table 2), and the stability constants for the competing cations (Table 4). Although it is recognized that Zn-binding properties of fish gills may change during exposure to Zn (28), the limited available data do not allow taking this into account in the chronic Zn BLM at present. Therefore, we adopted the $\log K_{\text{ZnBL}}$ and B_{max} values as reported by Alsop and Wood (28) for short-term exposures and used by Santore et al. (19) in the acute Zn BLM (i.e., $\log K_{\text{ZnBL}} = 5.5$, $B_{\text{max}} = 8.3 \text{ nmol of Zn g}^{-1} \text{ wet weight}$). Given the fact that mortality due to zinc in our chronic exposures mainly occurred during the initial stages of exposure, this is a plausible assumption for our BLM that predicts effects on survival. Additional studies are recommended to be able to take into account changes in gill Zn-binding into BLM modeling.

The following values were obtained (mean $\pm$ standard deviation, $\text{nmol of Zn g}^{-1} \text{ wet weight}$): 96-h $\text{LA}_{50} = 1.73 \pm 0.64$ ($n = 18$); 30-d $\text{LC}_{50} = 2.04 \pm 0.50$ ($n = 15$); and 30-d $\text{LA}_{10} = 0.77 \pm 0.31$ ($n = 14$). The 96-h and 30-d LA_{50} are not significantly different, which corroborates the fact that the toxic action of zinc is mainly of an acute nature. The 96-h LA_{50} is very similar to the values reported by Santore et al.

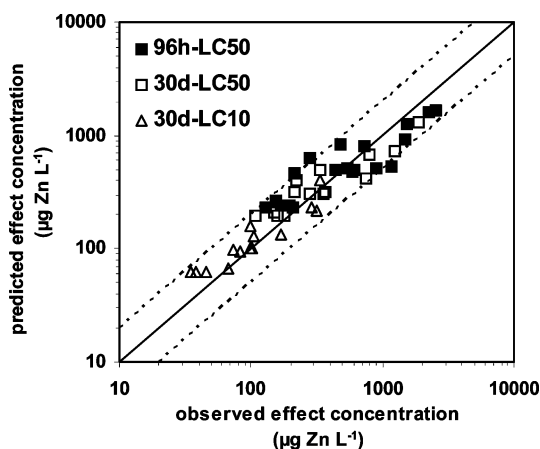


FIGURE 2. Observed vs predicted effect concentrations of zinc for rainbow trout. Predictions were performed using the BLM constants reported in Table 4. The bold line indicates a perfect match between observation and prediction, the dashed lines indicates a factor of 2 error between observation and prediction.

(19) (i.e., 0.63 and 2.29 nmol of Zn g⁻¹ wet weight for juveniles and adult rainbow trout, respectively).

Figure 2 depicts the predictive capacity of the developed acute and chronic Zn BLM. Except for two 96-h LC₅₀ values, all effect concentrations were predicted within a factor of 2 of the observed values. Hence, the observed variability of toxicity expressed as dissolved effect concentrations (factor of 10–30) is drastically reduced using the BLM. Chronic toxicity testing with spiked natural surface waters is ongoing in our laboratory as part of the ongoing evaluation of the predictive capacity of the BLM under field conditions.

Extrapolation of Study Results and BLM to Other Fish Species. In the previous sections, the development of a BLM that can predict chronic Zn toxicity to rainbow trout was presented. The ultimate aim of the BLM is to take into account bioavailability in the risk assessment and criteria setting of Zn and its substances in freshwater. However, the chronic BLM has been developed with only one fish species (rainbow trout), one life stage (swim fry), and one end point (survival). Regulatory action toward toxicants, however, should be aimed at protecting a certain percentage of species present in an ecosystem (typically 95%), meaning that all life stages must be protected and not only against effects on survival but also against effects on other relevant chronic end points such as growth, reproduction, and hatchability.

In this context, it may be considered an advantage that the swim fry stage of rainbow trout used in this study seems to be the life stage most sensitive to zinc. Indeed, Sinley et al. (17) observed that the swim fry stage of rainbow trout (NOEC_{s,swim fry} = 140 µg of Zn L⁻¹, LOEC_{s,swim fry} = 260 µg of Zn L⁻¹) was more sensitive to zinc than the sac fry and the egg/embryo stage (LOEC_{s,sac fry} and LOEC_{s,egg} > 547 µg of Zn L⁻¹) in an exposure initiated with eyed eggs. Additionally, when swim fry were not exposed as eggs/embryos, a higher sensitivity was noted (NOEC_{s,swim fry} = 36 µg of Zn L⁻¹, LOEC_{s,swim fry} = 71 µg of Zn L⁻¹), which also suggests that the swim fry stage is the most sensitive to zinc. The swim fry stage of brook trout (*Salvelinus fontinalis*) has also been demonstrated to be more sensitive to zinc than other life stages (Table 1) (10). Additional research is recommended to investigate if this is also true for fish species other than rainbow trout or brook trout.

A second advantage in the context of using the developed BLM in a risk assessment framework is that this model was developed for the most sensitive end point (i.e., survival). Indeed, in this study it was demonstrated that survival is a more sensitive end point than growth. This confirmed the

observations of Sinley et al. (17), who also indicated (qualitatively) that reproduction of rainbow trout surviving a chronic Zn exposure was not adversely affected (Table 1). Additionally, it was demonstrated, both in the present study and by Sinley et al. (17), that mortality mostly occurs in the first few days of the exposure to Zn (Table 1), suggesting a mainly acute mode of action and mortality being the driver of chronic toxicity test results. All this, together with survival being the most sensitive end point, indicates that rainbow trout that survive the first few days of an exposure to sublethal concentrations of zinc may acclimate without any major long-term costs involved.

However, despite these advantages, the following additional issues may arise when extrapolating the observations of this study and the developed BLM to other species: (i) Does the mainly acute nature of Zn toxicity suggested in the present study also apply to other fish species, and is there a mechanistic explanation? (ii) Are bioavailability-modifying effects (i.e., cation competition) similar for other species and other life stages (e.g., the embryo/egg stage and the sac fry stage), and is there a mechanistic explanation for the importance of competition by the different cations? The first question can be divided in two sub-questions: is the survival end point also the most sensitive one as compared to other chronic end points and does mortality also occur in the initial stages of the exposure? The second question can be addressed both qualitatively (comparing trends) and quantitatively (comparing stability constants). The answer to these questions is of crucial importance to determine how safely the BLM can be used to take into account bioavailability of zinc for predicting effects to fish populations under long-term exposures.

These questions will be addressed below based on existing literature data. In literature, to our knowledge, chronic toxic effects of zinc have been investigated with seven fish species in the past 30 years (i.e., 1973–2003): brook trout (*Salvelinus fontinalis*, 10), fathead minnows (*P. promelas*, 11–13), flagfish (*Jordanella floridae*, 14), humble minnow (*Phoxinus phoxinus*, 15), mottled sculpin (*Cottus bairdi*, 16), rainbow trout (*O. mykiss*, 17 and the present study), and zebrafish (*Danio rerio*, 18) (Table 1). Toxicity studies were considered chronic if they lasted longer than 96 h and if end points other than survival were reported. The data reported in these studies have been used to compare relative sensitivities of end points and to test the hypothesis of a mainly acute mode of action (Table 1).

For comparing survival with other end points in an objective manner, a scoring system was adopted that takes into account both statistical analyses and concentration–response relationships reported in the above-mentioned studies (Table 1). A score close to 1 indicates that survival is very likely to be the most sensitive end point, a score close to 0 indicates that another end point is more sensitive, a score close to 0.5 suggests more or less equal sensitivity of survival and another end point. On the basis of this scoring system, survival can be considered at least equally as sensitive as any other chronic end point measured (hatching, reproduction, growth), except for fathead minnows (Table 1). This analysis could not be performed for zebrafish. For rainbow trout, brook trout, humble minnow, and mottled sculpin a score ≥ 0.75 was obtained, indicating that survival was the most sensitive end point. For flagfish, a score of 0.5 indicated that survival was more or less equally sensitive as growth, whereas it was more sensitive than reproduction and hatchability. Three chronic studies were available on fathead minnows (*P. promelas*). In 5–7-d tests with sac fry survival and growth can be considered more or less equally sensitive end points with a score of 0.36 and 1 (11, 12). However, in 32-d tests, survival was more sensitive than growth. This corroborates the findings of Hogstrand et al. (27) for rainbow

trout: growth effects occur early in the exposure (first 14 d) and are compensated for within a reasonable short time (~30 d). Despite this compensation of growth effects, Benoit and Holcombe (13) demonstrated that reproduction was more sensitive an end point than survival in a life cycle exposure of fathead minnows initiated with eggs. Reproduction was significantly reduced at concentrations lower than those affecting survival ($LOEC_r = 145 \mu\text{g of Zn L}^{-1}$; $LOEC_s = 295 \mu\text{g of Zn L}^{-1}$). However, differences between $LOEC_s$ and $LOEC_r$ were still relatively small (i.e., factor 2). Overall, for all species considered it must be concluded that the sensitivity of the survival end point is at least close to the sensitivity of other end points, with the majority of the data suggesting that survival is the most sensitive end point. Additionally, mortality of zinc exposed fish usually occurs in the initial stages of the exposure. Available data for five species indicate that mortality only occurs in the first 2 weeks of exposure. In general, mortality ceased somewhere between day 4 and day 14 (Table 1).

The above-mentioned arguments clearly suggest that the toxic mode of action of zinc is mainly of an acute nature. This mode of action in fish has been identified as an ionoregulatory disturbance resulting in decreased Ca levels in the blood, also known as hypocalcaemia (27), which eventually result in the death of the fish. The latter may result from both competitive and noncompetitive inhibition of Ca uptake via the apical Ca channel and from noncompetitive interference of Zn with the basolateral Ca^{2+} ATPase (27, 29). Hogstrand et al. (27) demonstrated a restoration of plasma Ca levels within 1 week during a 30-d exposure of rainbow trout to zinc after Ca was initially reduced in the earliest stages of the exposure. The latter was the result of an acclimation response characterized by limiting apical influx of Zn^{2+} and a restoration of apical and basolateral Ca^{2+} influx. This rapid restoration of normal Ca levels in the plasma may explain why zinc-induced mortality only occurs in the initial stages of the exposure for most fish species (Table 1).

The acclimation process is initially associated with an elevated metabolic cost (i.e. increased protein turnover) resulting in a temporarily stalled growth around the 14th day of exposure (27). However, these temporary effects were compensated within 1 month of exposure (27). This also corroborates our data and the available data for fathead minnow. Indeed, with fathead minnows growth effects were observed in 5–7-d exposures but not in a 32-d exposure (11) (Table 1).

Alsop et al. (20) demonstrated that rainbow trout exposed for 30 d to sublethal zinc did not exhibit much of a detectable long-term cost associated with this acclimation process. This may explain why survival of rainbow trout is a more sensitive end point than growth (this study; 17) and reproduction (17) in rainbow trout (Table 1). Hence, bioavailability models for rainbow trout based on survival data are most likely also suitable for protecting rainbow trout against effects on growth and reproduction. It may also explain why for the majority of the literature data survival is at least an equally sensitive end point as any other chronic end point. For fathead minnows, however, reproduction was reduced at a concentration that did not affect survival. We hypothesize that the acclimation cost for this species was just large enough to have substantial reproductive consequences and that the impaired Ca balance is also, as for other species, the basis of chronic toxic effects.

For the BLM to be applicable to other fish species, the bioavailability relations observed for rainbow trout (i.e., competition reactions at the biotic ligand) should also hold for other fish species and for life stages other than the swim fry stage (at least qualitatively). Given the relative importance of Ca competition (factor 12 variability) as compared to the effects of Mg, Na, and pH (factor 2–3 variability), the Ca

effect should be evaluated first. Since the mode of action of zinc is mainly acute, a comparison of the effects of competing cations using both chronic and acute toxicity data is valid.

The importance of Ca in reducing Zn toxicity is logical because zinc toxicity mainly occurs through interference with Ca transport systems and because Ca and Zn share uptake pathways in the gill (29). Similarly, Na has been shown to be more important than Ca and Mg in reducing chronic toxicity of copper, a metal interfering with Na homeostasis (8). The greater importance of Ca competition compared to Mg and Na has previously also been demonstrated for acute and chronic zinc toxicity to *D. magna* (ref 3 and Heijerick et al., unpublished data) and for Zn uptake and acute toxicity in rainbow trout (20). Similarly Ca protects *Hyallela azteca* more than Mg and Na against toxicity of Cd (30), a metal that also acts on Ca homeostasis (31). Alsop and Wood (20) explain the lower affinity of Mg and Na (as compared to Ca) for zinc binding sites on the gill as the result of a nonspecific competition between Mg, Na, and Zn for anionic sites on the gill. Additional studies with Cd indicate that Ca is also a very important competitor for Cd binding to fathead minnow gills (32), Cd uptake in the humble minnow *P. phoxinus* (33) and for Cd toxicity to zebrafish embryos and sac fry (34). The study with *P. phoxinus* demonstrates that Ca competition also occurs in species other than rainbow trout and fathead minnows (33). The study with zebrafish (34) suggests that the competition with Ca not only occurs in juvenile and adult life stages but also in the earliest life stages.

Concluding, protective Ca competition effects with Zn or Cd have been conclusively demonstrated for different fish species (and invertebrates) and different life stages. Hence, Ca competition must be considered a ubiquitous concept for all fish and most likely for all life stages.

Contrary to effects of Ca on Zn (or Cd) toxicity or uptake, to our knowledge no other reports are available on the effects of Mg or Na on Zn (or Cd) toxicity to fish species other than rainbow trout. The protective effect of Mg against zinc toxicity reported in our study has also been observed for Cd toxicity to rainbow trout in both acute (96 h) and chronic exposures (100 d) (35). On the contrary, the competition effect of H^{+} on zinc toxicity observed in this study has also been suggested earlier with both rainbow trout and fathead minnows (19).

The effects reported in this study can be compared with other studies based on stability constants for competing cations (Table 4). Next to the constants derived in the present study, constants are also reported in the acute Zn BLM studies with rainbow trout and fathead minnow (19) and in the acute Zn BLM studies with *D. magna* (3). Although the stability constants for acute zinc toxicity to rainbow trout derived in the present study tend to be a little higher than those for acute zinc toxicity to *Daphnia* (~0.5 log units), the order of importance $\text{Ca} > \text{Mg} > \text{Na}$ also seems to hold for this species.

When the stability constants for acute zinc toxicity to rainbow trout obtained in the present study are compared with those described by Santore et al. (19), it appears that Mg and Na competition were not taken into account in the model of Santore et al. (19), simply because there was no evidence for Mg and Na protecting against zinc toxicity to fish species at the time those authors developed their Zn BLM. The Cd BLM, however, developed by the same authors (19), with some stability constants based on ref 32, does contain constants for Mg and Na that were very similar to our constants, especially for acute exposures. More importantly, the model developed by Santore et al. (19) has a stability constant for Ca that is about 1 order of magnitude higher than the one we have derived for our acute BLM. Perhaps, this may point to the importance of food during acute exposures. Santore et al. (19) developed the acute zinc BLM based on toxicity data that were mostly obtained in the absence of food, whereas in the present study food was

supplied to the fish. It could be hypothesized that fish may be able to compensate the Ca loss via the exposed gill through an increased Ca uptake from the diet. Indeed, it has been demonstrated that feeding can protect against ionoregulatory disturbance (36, 37). This may result in less apparent protective effect of waterborne Ca on zinc toxicity.

Finally, the Ca constant obtained in our study ($\log K_{CaBL} = 3.6$ and 3.8 for chronic and acute exposure, respectively) clearly confirms the importance of competition of Ca and Zn at the apical Ca channel in modifying acute and chronic zinc toxicity. Indeed, the Michaelis–Menten half-saturation constant of Ca for this channel is about $100\text{--}200\ \mu\text{mol L}^{-1}$ (27), which agrees with a $\log K_{CaBL} = 3.7\text{--}4.0$ (the inverse of the half-saturation constant), which is very similar to the values derived based on toxicity data in the present study. This again demonstrates the mechanism and importance of Ca for reducing zinc toxicity. The fact that the epithelial Ca channel is highly conserved among vertebrates (Christer Hogstrand, personal communication) suggests that the Ca competition concept is a ubiquitous phenomenon in fish and that the values of $\log K_{CaBL}$ will probably also be very similar for most fish species. The similarity of constants in fish and *Daphnia* suggests that this may also apply to invertebrate taxa.

Since the Ca effect is by far the most important modifier of chronic Zn toxicity to fish (and probably also to other species) and since the uncertainty about this effect is relatively small (both qualitatively and quantitatively), the remaining uncertainty about extrapolating the BLM concept to other fish species or life stages is considered relatively small as compared to its capacity to reduce toxicity variability due to bioavailability. However, as dissolved organic matter has been shown to be an important modifier of zinc toxicity (3, 4, 7, 19) and as this effect has not been investigated in the present study, caution should still be employed to use the BLM developed in this study for predictions of zinc toxicity in natural surface waters.

Acknowledgments

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Literature Cited

- De Schampelaere, K. A. C.; Janssen, C. R. *Environ. Sci. Technol.* **2002**, *36*, 48–54.
- De Schampelaere, K. A. C.; Heijerick, D. G.; Janssen, C. R. *Comp. Biochem. Physiol.* **2002**, *133C*, 243–258.
- Heijerick, D. G.; De Schampelaere, K. A. C.; Janssen, C. R. *Environ. Toxicol. Chem.* **2002**, *21*, 1309–1315.
- Heijerick, D. G.; Janssen, C. R.; De Coen W. M. *Arch. Environ. Contam. Toxicol.* **2003**, *44*, 210–217.
- Paquin, P. R.; Gorsuch, J. W.; Apte, S.; Batley, G. E.; Bowles, K. C.; Campbell, P. G. C.; Delos, C. G.; Di Toro, D. M.; Dwyer, R. I.; Galvez, F.; Gensemer, R. W.; Goss, G. G.; Hogstrand, C.; Janssen, C. R.; McGeer, J. C.; Naddy, R. B.; Playle, R. C.; Santore, R. C.; Schneider, U.; Stubblefield, W. A.; Wood, C. M.; Wu, K. B. *Comp. Biochem. Physiol.* **2002**, *133C*, 3–36.
- Di Toro, D. M.; Allen, H. E.; Bergman, H.; Meyer, J. S.; Paquin, P. R.; Santore, C. S. *Environ. Toxicol. Chem.* **2001**, *20*, 2383–2396.
- Heijerick, D. G.; De Schampelaere, K. A. C.; Janssen, C. R. *Comput. Biochem. Physiol.* **2002**, *133C*, 207–218.
- De Schampelaere, K. A. C.; Janssen, C. R. *Environ. Toxicol. Chem.* **2004**, *23*, 1365–1375.
- European Chemicals Bureau Newsletter Issue 3; European Chemicals Bureau, Institute of Health and Consumer Protection, Joint Research Centre, European Commission, 2003.
- Holcombe, G. W.; Benoit, D. A.; Leonard, E. N. *Trans. Am. Fish Soc.* **1979**, *108*, 76–87.
- Norberg-King, T. J. *Environ. Toxicol. Chem.* **1989**, *8*, 1075–1089.
- Norberg, T. J.; Mount, D. J. *Environ. Toxicol. Chem.* **1985**, *4*, 711–718.
- Benoit, D. A.; Holcombe, G. W. *J. Fish Biol.* **1978**, *13*, 701–708.
- Spehar, R. L. *J. Fish Res. Board Can.* **1976**, *33*, 1939–1945.
- Bengtsson, B. *Oikos* **1974**, *25*, 370–373.
- Woodling, J.; Brinkman, S.; Albeke, S. *Environ. Toxicol. Chem.* **2002**, *21*, 1922–1926.
- Sinley, J. R.; Goettl, J. P., Jr.; Davies, P. H. *Bull. Environ. Contam. Toxicol.* **1974**, *12*, 193–201.
- Dave, G.; Damgaard, B.; Grande, M.; Martelin, J. E.; Rosander, B.; Viktor, T. *Environ. Toxicol. Chem.* **1987**, *6*, 61–71.
- Santore, R. C.; Mathew, R.; Paquin, P. R.; Di Toro, D. *Comp. Biochem. Physiol.* **2002**, *133C*, 271–285.
- Alsop, D. H.; Wood, C. M. *Can. J. Fish. Aquat. Sci.* **1999**, *56*, 2112–2119.
- Guideline for Testing of Chemicals*, No. 215; OECD: Paris; 2000.
- Van Ewijk, P. H.; Hoekstra, J. A. *Ecotoxicol. Environ. Saf.* **1993**, *25*, 25–32.
- Bruce, R. D.; Versteeg, D. J. *Environ. Toxicol. Chem.* **1992**, *11*, 1485–1494.
- Levenberg, K. Q. *J. Appl. Math.* **1944**, *2*, 164–168.
- Marquardt, D. W. *J. Soc. Ind. Appl. Math.* **1963**, *11*, 431–441.
- Martell, A. E.; Smith, R. M.; Motekaitis, R. J. *Critical Stability Constants of Metal Complexes Database, version 4.0*. NIST Standard Reference Database 46; National Institute of Standards and Technology: Gaithersburg, MD, 1997.
- Hogstrand, C.; Reid, S. D.; Wood, C. M. *J. Exp. Biol.* **1995**, *198*, 337–348.
- Alsop, D. H.; Wood, C. M. *Environ. Toxicol. Chem.* **2000**, *17*, 1911–1918.
- Hogstrand, C.; Verbost, P. M.; Wendelaar Bonga, S. E.; Wood, C. M. *Am. J. Physiol. R* **1996**, *270*, 1141–1147.
- Jackson, B. P.; Lasier, P. J.; Miller, W. P.; Winger, P. W. *Bull. Environ. Contam. Toxicol.* **2000**, *64*, 279–286.
- Verbost, P. M.; Van Rooij, J.; Flik, G.; Lock, R. A. C.; Wendelaar Bonga, S. E. *J. Exp. Biol.* **1989**, *145*, 185–197.
- Playle, R. C.; Dixon, D. G.; Burnison, K. *Can. J. Fish. Aquat. Sci.* **1993**, *50*, 2678–2687.
- Wicklund, A.; Runn, P. *Aquat. Toxicol.* **1988**, *13*, 109–122.
- Meinelt, T.; Playle, R. C.; Pietrock, M.; Burnison, B. K.; Wienke, A.; Steinberg, C. E. W. *Aquat. Toxicol.* **2001**, *54*, 205–215.
- Davies, P. H.; Gorman, W. C.; Carlson, C. A.; Brinkman, S. F. *Chem. Speciation Bioavailability* **1993**, *5*, 67–77.
- Dockray, J. J.; Reid, S. D.; Wood, C. M. *Can. J. Fish. Aquat. Sci.* **1996**, *53*, 2752–2763.
- Baldiasserotto, B.; Kamunde, C.; Matsuo, A.; Wood, C. M. *Aquat. Toxicol.* **2004**, *67*, 57–73.

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